



# Ratnesh Kumar Singh

Electronics & Communication  
Engineer

Networking • AI Automation • UI/UX

## CONTACT

- +91 70049 37129
- kumarsinghratnesh3@gmail.com
- West Bengal, India
- portfolio-omega.vercel.app
- tinywebs.site/6AsQaP

## SKILLS & TOOLS

### NETWORKING & IT

TCP/IP, DNS, Routing & Switching,  
Cybersecurity

### PROGRAMMING

C, C++, Python, HTML, CSS, API  
Integration

### AI & AUTOMATION

n8n, Google AI Studio, OpenAI, AI  
Agents

### EMBEDDED & ENG.

Arduino, MATLAB, Tinkercad, Verilog,  
ArcGIS, HFSS Simulation

### UI/UX & DESIGN

Figma, Wireframing, Prototyping, Canva

## LANGUAGES

English  
Hindi  
Bengali

## INTERESTS

Photography, Sketching, Audio & Video  
Editing, Tech Trends

## PROFILE SUMMARY

Motivated Electronics & Communication Engineering graduate (2026) with a versatile skillset spanning networking, IT infrastructure, AI automation, and UI/UX design. Experienced in developing embedded hardware prototypes, designing user interfaces in Figma, and programming in Python/C++. Strong understanding of networking concepts (TCP/IP, routing), V2X communication, and workflow automation. Eager to contribute to technology-driven roles across networking, software development, or systems integration. Open to relocation across India.

## KEY PROJECTS

### Adaptive Smart Antenna (V2X) (V2X • ECE Hardware)

Designed and developed an adaptive smart antenna system prototype to improve wireless communication reliability in high-mobility vehicular environments. Implemented beamforming algorithms that dynamically direct signals toward moving vehicles, helping reduce interference and maintain stable connectivity. Supporting emerging V2X (Vehicle-to-Everything) standards.

### AI Assistant (Python • APIs • Conversational AI)

Created an intelligent assistant using Python and API integrations. Implemented features like natural language command understanding, conversational response generation, and real-time information retrieval, demonstrating client-server communication and data handling skills.

### Smart Parking System (Arduino • Ultrasonic Sensors • IoT)

Designed an IoT prototype using Arduino and ultrasonic sensors to detect available parking spaces in real-time, demonstrating practical sensor integration and network communication.

### UI/UX Design & Portfolio Website (Figma • HTML • CSS • Vercel)

Designed multi-screen UI layouts and interactive prototypes in Figma, enforcing accessibility and typography guidelines. Developed and deployed a clean, responsive portfolio website on Vercel.

## EDUCATION

| Qual.    | Institution                             | Board/University  | Year        | Specialization |
|----------|-----------------------------------------|-------------------|-------------|----------------|
| B.Tech   | Swami Vivekananda Inst. of Sci. & Tech. | MAKAUT (W.B.U.T.) | 2022 - 2026 | ECE            |
| 12th Std | P.B.S College                           | B.S.E.B           | 2020        | Science (PCM)  |
| 10th Std | Vidyanjali High School (H.S.)           | I.G.C.S.E         | 2018        | Science        |

## CERTIFICATIONS & TRAINING

### Python Programming

1-Month Course

### Electric Vehicle Service Tech.

1-Month Training

### C Language Programming

1-Month Course

### GIS (Geographic Info. Systems)

2-Weeks Training

## EXTRA-CURRICULAR ACTIVITIES

- HAM Radio:** Workshop on Innovation & General Awareness (2025).
- BSNL Telecom:** Completed a one-day industrial visit at BSNL Telecom (2025).
- Cyber Security:** Workshop 'Empower Yourself With Cyber Awareness' (2025).

